```
    result.extend(left[i:])
    result.extend(right[j:])
    return result
```

Explanation:

- `merge_sort()` splits the list into two halves, recursively sorts them, and merges them using the `merge()` function.
- The `merge()` function combines two sorted lists into one.

5. Quick Sort

Quick Sort is another divide-and-conquer algorithm. It picks a pivot element, partitions the list into elements smaller and greater than the pivot, and then recursively sorts each partition.

```
def quick_sort(data):
    if len(data) <= 1:
        return data
    pivot = data[len(data) // 2]
    left = [x for x in data if x < pivot]
    middle = [x for x in data if x == pivot]
    right = [x for x in data if x > pivot]
    return quick_sort(left) + middle + quick_sort(right)
```